

Question number	Answer	Additional guidance	Mark
5(a)	C is the correct answer • a high glomerular filtration does not enable the camel to reduce water loss • a slow release of ADH does not enable the camel to reduce water loss		Computer (1)

Question number	Answer	Additional guidance	Mark
5(b)(i)	A calculation with the following steps • calculation of water use each day for camel and buffalo (1) • correct difference in dm³ to two significant figures(1)	$(57 \times 850) = 48450 \text{ cm}^3$ $(150 \times 697) = 104550 \text{ cm}^3$ $(104550 - 48450) \div 1000 = 56 \text{ (dm}^3\text{)}$ 56.1= 1 mark Power of 10 error 1 mark eg 561 150-57=93 and then dividing by 1000 to get 0.093 = 1 mark	Graduate (2)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An answer that includes the following points: • credit one adaptation / environmental condition (1) • credit second adaptation / environmental condition (1)	accept different habitat accept different environmental condition	Expert (2)

Question number	Answer	Additional guidance	Mark
5(c)	An answer that includes three of the following points: • (overall) both fresh water and sea water increased (mean) mass (of kangaroo rats) (1) • fresh water increased (mean) mass (of kangaroo rats) more than sea water • credit any calculated change in mass over days/ or daily change (1)	can piece together from separate descriptions of data accept fresh water caused the highest increase in (mean) mass e.g. 6 g difference between fresh and no water / 0.5 g difference between fresh and salt water / decrease of {0.2/0.6/1.8} g after 2 days / increase of {0.5 / 1} g after 16 days Fresh 0.83% and sea 0.42% daily	Expert (3)

Question number	Answer	Additional guidance	Mark
5(d)	An answer that includes two of the following points: • (nephron has) long(er) loop of Henle (1) • higher concentration of ions in the medulla (1) • (kidneys/tubule/ DCT and collecting duct/ loop of Henle) reabsorb {large quantities of / more } water (1)	accept reference to counter current multiplier accept ADH results in increased absorption of water accept increased permeability to water in DCT and collecting duct accept ADH causes insertion of {water channels / aquaporins} in DCT and collecting duct	Expert (2)

